Avinash Bhojanapalli

Machine Learning Engineer

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EXPERIENCE

dCortex | *Founding Machine Learning Engineer* May 2026 - Sep 2026

- Cut last-minute coverage scrambles ~40% by building a live 72-hour risk model that scores 8,000+ crew-duty pairs per window in under 100ms.
- Cut override diagnosis ~70% from full-slate review to seconds by building counterfactual analysis across 400+ live decisions.
- Built a chat agent that writes and runs SQL on the fly so controllers can fetch crew identity, reserve status, and legality details in plain language, cutting lookup time **~80%** (minutes of Airline Management System clicks to seconds).

Bonsai Lab RIT | *Machine Learning Engineer* Aug 2025 - Apr 2026

- Reduced experiment iteration time by 60% by architecting a modular PyTorch self-supervised learning pipeline for rapid representation comparisons.
- Optimized data loading pipelines with pre-fetching and caching, which increased GPU utilization and reduced training time.
- Implemented automated experiment tracking and artifact versioning, ensuring reproducibility across distributed training runs and standardizing metric reporting for team-wide collaboration.

Incribo | *MLOps Engineer Intern* Jan 2024 - Jul 2024

- Designed and deployed a fault-tolerant microservices architecture for multi-agent AI systems on Kubernetes and Docker, using RabbitMQ for robust asynchronous messaging, which improved system reliability and enabled seamless scaling of AI agents
- Reduced GPU memory usage by 15% and improved inference latency by optimizing LLM serving with gradient checkpointing and dynamic batching.
- Developed a **Go-based** CLI tool to automate container orchestration and deployment workflows, reducing deployment friction and ensuring consistent environments across development and production.
- Integrated **Synthetic Data Generation** workflows using neuroevolution algorithms (CPPN-NEAT), scaling data generation rates by **15%** to support robust model training and validation.

SKILLS

- **Languages**: Python, C++, Go, Rust, SQL, Bash
- **Backend & Systems**: RESTful APIs, Microservices, FastAPI, Linux, Git
- **MLOps & Cloud**: Docker, Kubernetes, AWS (SageMaker, Lambda, ECR), CI/CD (GitHub Actions), MLflow, GCP (Vertex AI), Grafana
- **Machine Learning**: PyTorch, TensorFlow, Scikit-learn, Pandas, HuggingFace, Langchain, ONNX
- **Data & Databases**: PostgreSQL, MongoDB, Redis, ChromaDB (Vector DB), Apache Kafka, Neo4j, SQLite

EDUCATION

Rochester Institute of Technology | *MS Data Science* (GPA: 3.9/4.0) **Rochester, NY Aug 2024 - Apr 2026**

IIIT Sri City | *BTech Computer Science and Engineering* **India Dec 2020 - May 2024**

KEY PROJECTS

Crosstem: High-Performance NLP Library | Rust, Python, PyO3, CI/CD PyPI

- Engineered a multilingual NLP stemming library with a Rust-accelerated core (PyO3) and Python API, delivering 2.7-2.9x speedup over the Python fallback and 15x higher throughput than Porter on benchmark workloads.
- Built a graph-based morphological stemming pipeline (BFS + productivity-threshold scoring) that resolves overstemming and correctly maps across part-of-speech boundaries.
- Shipped and maintained the library on PyPI with parity-tested Rust/Python backends, production packaging, and end-to-end documentation for installation, API usage, and performance validation.

Ledgerline: Vectorless RAG Intelligence Engine | Go, Python, Kafka, Redis Github

- Engineered an event-driven, vectorless RAG microservices stack, replacing traditional chunking with hierarchical tree navigation to eliminate context loss across complex 200+ page financial filings.
- Architected a high-throughput API Gateway in Go (Fiber) featuring WebSocket streaming for real-time AI reasoning, integrating Redis caching to hot-load document trees and maintain query latency strictly under 3 seconds.
- Orchestrated asynchronous Kafka pipelines for reliable service handoff, integrating an automated LLM-as-a-judge observability system backed by PostgreSQL to continuously evaluate response telemetry and rigorously prevent hallucinations.

Topological Arbitrage Engine: Market-Neutral Quant Strategy | Python, Pandas, SciPy, Flask Github

- Developed a statistical arbitrage trading engine applying Graph Laplacian diffusion to CAPM residuals, isolating idiosyncratic alpha to exploit structural mean-reversion across a dynamic 80-stock universe.
- Architected a rigorous walk-forward validation pipeline that proved the raw topological signal thrives during market volatility, achieving a 42.5% annualized return and a 1.99 Sharpe ratio on strictly out-of-sample 2025-2026 data.
- Deployed a fully automated live paper-trading infrastructure featuring daily execution, equal-weight position sizing, and state management, integrated with a local web dashboard for real-time portfolio telemetry.

PUBLICATIONS

- A Bhojanapalli, et al..A CALIPSO Observation Based 3-Dimensional Tropospheric Aerosol Classification Model Over the Indian City Delhi.*In 2023 IEEE International Geoscience and Remote Sensing Symposium, pp. 305-308. IEEE. 16-21 July 2023, Pasadena, CA, USA. DOI:10.1109/IGARSS52108.2023.10282609*
- Z Lyu, et al..Evolving RNNs for Stock Forecasting: A Low Parameter Efficient Alternative to Transformers.*In International Conference on the Applications of Evolutionary Computation (Part of EvoStar), pp. 36-54. Springer, Cham. 17 April 2025, Trieste, Italy. DOI:10.1007/978-3-031-90065-5_3*